

Question

Write a Java program to demonstrate the handling of `NoSuchMethodException` using try-catch.

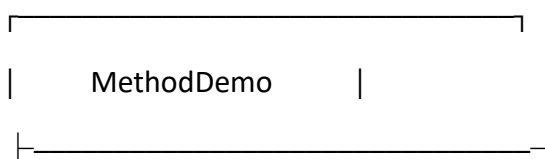
Steps for Execution

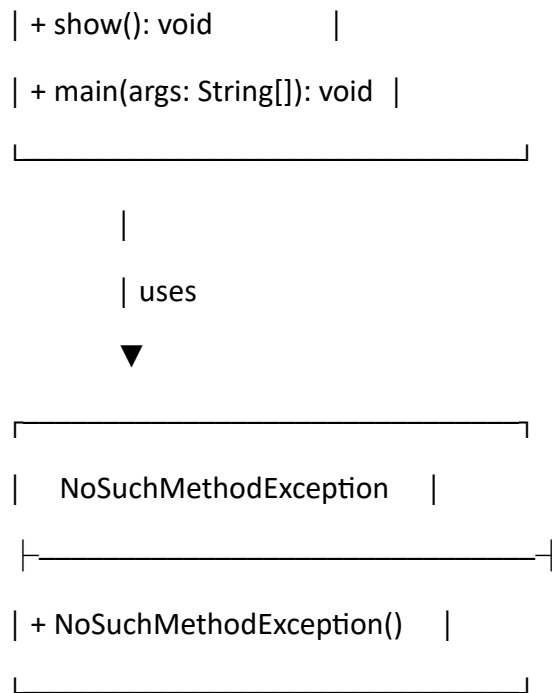
1. Create a package named `javapractice`.
2. Create a class named `MethodDemo`.
3. Define a method named `show()`.
4. Create an object of the `MethodDemo` class inside the try block.
5. Use `getClass().getMethod("display")` to search for a method named `display`.
6. Since the class does not contain a method named `display`, Java throws `NoSuchMethodException`.
7. The catch block catches the exception.
8. Display "Exception: Method not found".
9. Stop the program.

Algorithm

1. **Start**
2. Define class `MethodDemo`.
3. Define a public method `show()` that prints "Hello".
4. Enter the `main()` method.
5. Create an object of `MethodDemo`.
6. Try to find the method `display` using reflection.
7. If the method is not found, catch `NoSuchMethodException`.
8. Display "Exception: Method not found".
9. **Stop**

UML Class Diagram





Code

```
package javapractice;

public class MethodDemo {
    public void show() {
        System.out.println("Hello");
    }

    public static void main(String[] args) {
        try {
            MethodDemo obj = new MethodDemo();
            obj.getClass().getMethod("display");
        }
        catch (NoSuchMethodException e) {
            System.out.println("Exception: Method not found");
        }
    }
}
```

```
}
```

Output

Exception: Method not found

Result

The program successfully demonstrates `NoSuchMethodException` handling using a try-catch block. The exception occurs because the `MethodDemo` class contains `show()` but does not contain a method named `display`.